Exercise: 34.

Wilmott's bubbler experiment

Aim :

To determine rate of photosynthesis by Wilmott's bubbler

Requirements :

Wilmott's bubbler apparatus, Hydrilla twig, water.

Procedure :

1. Fill the bottle with water and insert Hydrilla twig into the wider part of the tube
2. Hydrilla plant should be cut inside the water to avoid entry of air bubbles
3. Fix the tube with jar which acts as water reservoir
4. Keep the apparatus in sunlight
5. Count the bubbles when they are in same size.
6. Repeat the experiment in different light intensity.

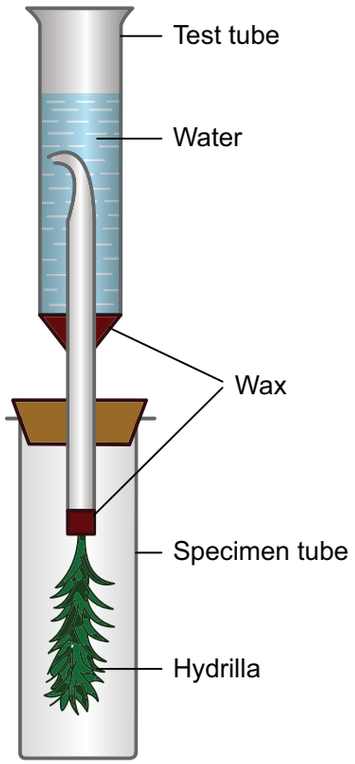


Figure 30 : Wilmott's bubbler

Observation :

When there is an increase in photosynthesis, bubble count also increased.

Inference :

Rate of photosynthesis increases with increase of light intensity is proved

Exercise: 35

Experiment to demonstrate the production of CO2 in aerobic respiration.

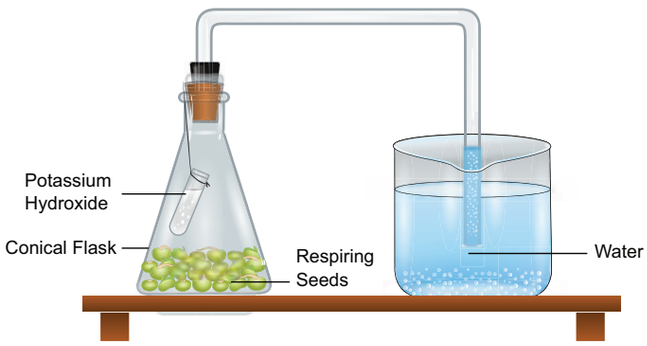


Figure 31: Demonstration of production of CO_2 during aerobic respiration

11th Botany_EM_Chapter 15.indd 370 11/01/2023 15:00:50